

Solutions to the Practice Final Exam (STAT 461)

(9) If $Y_i \sim \text{Uniform}[0, \theta]$, then the first population moment is

$$E(Y_i) = \frac{0 + \theta}{2} = \frac{\theta}{2}.$$

The method-of-moments estimator is obtained by solving

$$\frac{\theta}{2} = \bar{Y}.$$

Therefore,

$$\hat{\theta}_{\text{MOM}} = 2\bar{Y}.$$

Next, since

$$\text{Var}(Y_i) = \frac{\theta^2}{12},$$

the MOM estimator of θ^2 is obtained by solving

$$\frac{\theta^2}{12} = S^2,$$

where

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2$$

is the sample variance. Thus,

$$\hat{\theta}_{\text{MOM}}^2 = 12S^2.$$

(10) Let $X_1, X_2, \dots, X_n$ be i.i.d., each following a Weibull(3, β) density. The Weibull density is

$$f(x) = \frac{\alpha}{\beta^\alpha} x^{\alpha-1} e^{-(x/\beta)^\alpha}, \quad x > 0.$$

With $\alpha = 3$,

$$f(x) = \frac{3}{\beta^3} x^2 e^{-(x/\beta)^3}.$$

Therefore, the sample likelihood function is

$$L(\beta; x_1, \dots, x_n) = \prod_{i=1}^n \left(\frac{3}{\beta^3} x_i^2 e^{-(x_i/\beta)^3} \right) = \frac{3^n}{\beta^{3n}} \left(\prod_{i=1}^n x_i \right)^2 \exp \left(-\frac{\sum_{i=1}^n x_i^3}{\beta^3} \right).$$

The log-likelihood function is

$$\ell(\beta) = \ln L(\beta; x_1, \dots, x_n) = n \ln 3 - 3n \ln \beta + 2 \sum_{i=1}^n \ln(x_i) - \frac{\sum_{i=1}^n x_i^3}{\beta^3}.$$

Differentiate with respect to β and set equal to zero:

$$\frac{d\ell}{d\beta} = -\frac{3n}{\beta} + \frac{3 \sum_{i=1}^n x_i^3}{\beta^4} = 0.$$

Thus,

$$\frac{1}{\beta} \left(-3n + \frac{3 \sum_{i=1}^n x_i^3}{\beta^3} \right) = 0,$$

so

$$-3n + \frac{3 \sum_{i=1}^n x_i^3}{\beta^3} = 0.$$

Therefore,

$$\frac{\sum_{i=1}^n x_i^3}{\beta^3} = n, \quad \beta^3 = \frac{1}{n} \sum_{i=1}^n x_i^3,$$

and the maximum likelihood estimator is

$$\hat{\beta}_{\text{MLE}} = \sqrt[3]{\frac{\sum_{i=1}^n x_i^3}{n}}.$$

Checking the second derivative:

$$\frac{d^2 \ell}{d\beta^2} = \frac{3n}{\beta^2} - \frac{12 \sum_{i=1}^n x_i^3}{\beta^5} = \frac{1}{\beta^2} \left(3n - \frac{12 \sum_{i=1}^n x_i^3}{\beta^3} \right).$$

If we plug in $\hat{\beta}_{\text{MLE}}$ in place of β , then

$$\left. \frac{d^2 \ell}{d\beta^2} \right|_{\beta=\hat{\beta}_{\text{MLE}}} = \frac{1}{\hat{\beta}_{\text{MLE}}^2} (3n - 12n) < 0.$$

Therefore, this gives a maximum.

(11) A 99% confidence interval for the unknown population mean height μ is

$$\bar{x} \pm t_{0.005,15} \frac{s}{\sqrt{n}}.$$

Using $\bar{x} = 49$, $s = 6.4$, $n = 16$, and $t_{0.005,15} = 2.947$,

$$49 \pm (2.947) \frac{6.4}{\sqrt{16}} = 49 \pm 4.7152.$$

Thus, the 99% confidence interval is

$$(44.2848, 53.7152).$$

(12) We are testing

$$H_1 : \mu_1 > \mu_2 \quad \text{versus} \quad H_0 : \mu_1 \leq \mu_2,$$

that is,

$$H_1 : \mu_1 - \mu_2 > 0 \quad \text{versus} \quad H_0 : \mu_1 - \mu_2 \leq 0.$$

From the first sample of size $n_1 = 9$,

$$\bar{x} = 32.19, \quad s_1 = 4.34.$$

From the second sample of size $n_2 = 7$,

$$\bar{y} = 31.68, \quad s_2 = 4.56.$$

The test statistic is

$$t = \frac{\bar{x} - \bar{y} - 0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{32.19 - 31.68}{\sqrt{\frac{4.34^2}{9} + \frac{4.56^2}{7}}} = 0.2266.$$

It should be compared with $t_{0.05,6}$, because the degrees of freedom used here is the smaller of $n_1 - 1$ and $n_2 - 1$. Since

$$t_{0.05,6} = 1.943$$

and

$$0.2266 < 1.943,$$

we fail to reject H_0 . There is not sufficient evidence at the 5% significance level to claim that second graders who participate in regular sports have a higher mean score.

- (13) If the two unknown population standard deviations are known to be the same, $\sigma_1 = \sigma_2 = \sigma$, we use the pooled two-sample t test. The common population variance is estimated by

$$S_{\text{pool}}^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{(n_1 - 1) + (n_2 - 1)}.$$

Substituting the values,

$$S_{\text{pool}}^2 = \frac{(8)(4.34^2) + (6)(4.56^2)}{8 + 6} = 19.675.$$

The test statistic is

$$t = \frac{\bar{x} - \bar{y} - 0}{\sqrt{S_{\text{pool}}^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} = \frac{32.19 - 31.68}{\sqrt{19.675 \left(\frac{1}{9} + \frac{1}{7} \right)}} = 0.2281.$$

It is compared with

$$t_{0.05,14} = 1.761.$$

Since

$$0.2281 < 1.761,$$

we fail to reject H_0 .

- (14) For a 98% confidence interval,

$$z_{0.01} = 2.33.$$

The margin of error is

$$E = z_{0.01} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}.$$

- (a) For $n = 484$ and $\hat{p} = 98/484$,

$$E = (2.33) \sqrt{\frac{\frac{98}{484} \left(1 - \frac{98}{484} \right)}{484}} = 0.0425.$$

- (b) For $n = 1089$ and $\hat{p} = 655/1089$,

$$E = (2.33) \sqrt{\frac{\frac{655}{1089} \left(1 - \frac{655}{1089} \right)}{1089}} = 0.03457.$$

(c) For $n = 650$ and $\hat{p} = 540/650$,

$$E = (2.33) \sqrt{\frac{\frac{540}{650} \left(1 - \frac{540}{650}\right)}{650}} = 0.034267.$$

(15) Let X be the number of Ebola-infected individuals in a random sample of size 1000. Then

$$X \sim \text{Binomial}(n = 1000, p = 0.002).$$

Since

$$np = 1000(0.002) = 2,$$

this binomial distribution can be approximated by a Poisson distribution with mean $\mu = 2$:

$$X \approx \text{Poisson}(2).$$

Therefore,

$$P(X = 5) = e^{-2} \frac{2^5}{5!} = 0.0361.$$

Also,

$$P(X \leq 3) = P(X = 0, 1, 2, 3) = e^{-2} + e^{-2} \frac{2^1}{1!} + e^{-2} \frac{2^2}{2!} + e^{-2} \frac{2^3}{3!}.$$

Thus,

$$P(X \leq 3) = e^{-2} \left(1 + 2 + \frac{4}{2} + \frac{8}{6}\right) = 0.8571.$$

(16) Let R_T be the event that the transferred marble was red, and let R_D be the event that the marble drawn from the second bowl was red. Then

$$P(R_T) = \frac{6}{8} = \frac{3}{4}, \quad P(B_T) = \frac{2}{8} = \frac{1}{4}.$$

If a red marble was transferred, then the second bowl has 3 red marbles and 2 blue marbles, so

$$P(R_D | R_T) = \frac{3}{5}.$$

If a blue marble was transferred, then the second bowl has 2 red marbles and 3 blue marbles, so

$$P(R_D | B_T) = \frac{2}{5}.$$

Using Bayes' formula,

$$P(R_T | R_D) = \frac{P(R_T)P(R_D | R_T)}{P(R_T)P(R_D | R_T) + P(B_T)P(R_D | B_T)}.$$

Therefore,

$$P(R_T | R_D) = \frac{\left(\frac{3}{4}\right) \left(\frac{3}{5}\right)}{\left(\frac{3}{4}\right) \left(\frac{3}{5}\right) + \left(\frac{1}{4}\right) \left(\frac{2}{5}\right)} = \frac{9/20}{9/20 + 2/20} = \frac{9}{11}.$$